INCH-POUND
MIL-DTL-24643/28E
22 August 2002
SUPERSEDING
MIL-C-24643/28D
22 November 1994

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, TYPE LS1S50MU (INCLUDING VARIATION LS1S50MUS)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of MIL-DTL-24643 listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation.

This cable can be manufactured in three variations: LS1S50MU and LS1S50MUS (double overall shield).

REQUIREMENTS:

Qualification required.

Construction (non-watertight)

First -	Copper conductor, tin coated, ASTM B 286, size 22-7.
Second -	Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third -	Crosslinked polyethylene insulation, white. Nominal wall thickness 0.030 inch; wall thickness may vary from nominal as necessary, in order for completed cable to meet the specified electrical requirements.
Fourth -	Braided shield of AWG no. 36 tin coated copper. Braid angle of 20 to 25 degrees; minimum coverage of 90 percent.
Fifth -	Shield insulation of two polyester tapes, per NEMA FI 4, outer tape sealed, An alternate shield insulation may consist of one polyester tape, per NEMA FI 4, plus an extruded jacket of a suitable transparent material, minimum average thickness of the transparent material 0.003 inch. The standard identification code shall be applied by method 2 on the inner tape.
Sixth -	The required number of singles (see table I) cabled together with a lay not greater than 24 times the pitch diameter of the layer.
Seventh -	Binder tape, applied helically with overlap.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T00:04Z
Check the source to verify that this is the current version before use.

Construction variant for type LS1S50MU

Eighth - Cross-linked polyolefin jacket. Minimum average wall thickness 0.050 inch.

Construction variant for type LS1S50MUS

Eighth - Braided shield of tin coated copper. Minimum coverage of 85 percent.
Ninth - Spirally wrapped 0.002 inch thick polyethylene (polyester film) separator tape with 25 percent minimum overlap.
Tenth - Braided shield of tin coated copper. Minimum coverage of 85 percent.
Eleventh - Binder tape applied helically with overlap.
Twelfth - Cross-linked polyolefin jacket. Minimum average wall thickness 0.050 inch.

TABLE I. Details.

Military part no. 1/ M24643/28	Size 2/	Number of conductors	Diameter over shielded core (in. nom)	Overall diameter (inches)				Cold bending mandrel diameter maximum (inches)
				Type LS1S50MU		Type LS1S50MUS		
				min	max	min	max	
-01UN	16	16	0.655	0.760	0.825	0.820	0.885	11
-02UN	20	20	.760	0.835	0.905	0.895	0.965	12
-03UN	40	40	1.015	1.095	1.185	1.155	1.245	15
-04UN	70	70	1.380	1.465	1.555	1.525	1.615	20

1/ When double shield is required, see MIL-DTL-24643 for configuration of part number.
2/ Type LS1S50MU and LS1S50MUS.

EXAMINATION AND TESTS:

Basic electrical:	Requirements:
Conductor resistance - ohms/1000 feet at 25°C, maximum	17.54
Voltage withstand - volts, root mean square, minimum	
Conductor to shield	5000
Shield to shield (component shields only)	500
Shield to shield (type LS1S50MUS overall shields only)	200
Insulation resistance - megohms -1000 feet minimum	
Conductor to shield	2500
shield to shield (component)	100
shield to shield (LS1S50MUS only, overall shields)	100
Jacket flaws (LSMUS only)	No failure
Conductor and shield (LS1S50MUS only) continuity	No failure

Group A:	
Visual and dimensional	No failure
Characteristic impedance - at 1MHz, ohms	50 ± 2

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T00:04Z
Check the source to verify that this is the current version before use.

Group B:
- Cross-linked proof test (percent, maximum)
  - Insulation 50
  - Jacket (When tested at 200°C 50
- Attenuation - at 400 MHz, decibel (dB)/100 FT maximum 18
- Cold bending, cable - at -20 ± 2°C, (see table I for mandrel diameter) ..... No damage
- Tear - pounds per inch thickness, minimum (ASTM D 470) 35

Physicals (unaged)
- Insulation
  - Tensile strength - lb/in^2, minimum 1800
  - Elongation - percent, minimum 250
- Jacket (cable)
  - Tensile strength - lb/in^2, minimum 1300
  - Elongation - percent, minimum 160

Group C:

Physicals (aged)
- Insulation
  - Air oven
    - Tensile strength - percent of unaged, minimum 80
    - Elongation - percent of unaged minimum 80
- Jacket (cable)
  - Air oven
    - Tensile strength - percent of unaged, minimum 60
    - Elongation - percent of unaged minimum 60

  - Hot oil immersion
    - Tensile strength - percent of unaged, minimum 50
    - Elongation - percent of unaged minimum 50
- Shrinkage ..... No failure
- Heat distortion - percent of unaged, maximum 30
- Permanence of printing (jacket) - cycles, minimum 125
- Shield - conformance to material, construction and coverage ..... No failure
- Surface transfer impedance (type LS1S50MUS only)
  - Milliohms per meter, maximum 700
  - EMP response time - dB, minimum 60

Group D:
- Flame propagation (cable) ..... No failure

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

	Requirements:
Aging and compatibility (cable) (125 ± 5°C)	No failure
Abrasion resistance (jacket) - scrapes, minimum	75

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T00:04Z
Check the source to verify that this is the current version before use.

Acid gas equivalent - percent, maximum
- Jacket 2
- Fillers 2
- Insulation 18
- Shield insulation 2

Halogen content - percent, maximum
- Jacket 0.2
- Fillers 0.2

Immersion (jacket)
- Tensile strength - percent of unaged, minimum 50
- Elongation - percent of unaged, minimum 50

Smoke index, maximum
- Jacket 25
- Fillers 45
- Insulation 45

Toxicity index, maximum
- Jacket 5
- Fillers 5
- Insulation 1.5

Durometer (jacket) - (type A) hardness, minimum.....80

Weathering (jacket)..... No failure

UNIT ORDERING LENGTHS: 1000 feet nominal

Custodians:
Army - MI
Navy - SH

Preparing Activity:
Navy - SH
(Project 6145-2308-026)

Review Activities:
Army - AV, CR
Navy - CG, EC
DLA – CC

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T00:04Z
Check the source to verify that this is the current version before use.